HLD learning progression

1. Fundamentals and architecture

Levels 1 and 2

Start here

2. Communication and databases

Levels 3 and 4

3. Scalability and distributed systems

Levels 5 and 6

4. Reliability and architecture patterns

Levels 7 and 8

5. Machine Learning System Design

Level 9

6. Generative AI System Design

Level 10

7. Interview design problems

Practise progressively from beginner to advanced.

Level 1 — HLD Fundamentals

Beginner

First, understand the fundamental building blocks of software architecture.

No.	Topic	Concepts to cover
1	Introduction to HLD	Definition, purpose and importance
2	System Architecture	Components and architecture diagrams
3	Functional Requirements	What the system should do
4	Non-Functional Requirements	Scalability, reliability, security, performance
5	System Components	Frontend, backend, database, services
6	Client-Server Architecture	Client-server communication
7	Request-Response Lifecycle	How requests travel through a system
8	Monolithic Architecture	Single application architecture
9	Microservices Architecture	Multiple independent services
10	Architectural Trade-offs	Choosing between different approaches

Level 2 — Architectural Building Blocks

Beginner

Understand the individual components used to construct an application.

No.	Topic	Concepts to cover
1	Web Servers	Nginx, Apache
2	Application Servers	Uvicorn, Gunicorn
3	Reverse Proxy	Purpose and working
4	Load Balancers	L4 and L7 load balancing
5	API Gateway	Routing and request management

6	Database Servers	SQL and NoSQL
7	Cache Servers	Redis, Memcached
8	Message Brokers	Kafka, RabbitMQ
9	Object Storage	Amazon S3, file storage
10	Content Delivery Network	CDN and edge caching
11	DNS	Domain name resolution
12	Service Discovery	Finding available services

Level 3 — API and Communication Design

Beginner to Intermediate

Learn how different components communicate.

No.	Topic	Concepts to cover
1	API Fundamentals	API architecture
2	REST API Design	REST principles
3	HTTP Methods	GET, POST, PUT, PATCH, DELETE
4	HTTP Status Codes	2xx, 3xx, 4xx, 5xx
5	API Versioning	Managing API versions
6	API Authentication	API keys, JWT
7	Authorization	RBAC and permissions
8	Synchronous Communication	Request-response
9	Asynchronous Communication	Background processing
10	WebSockets	Real-time communication
11	gRPC	Inter-service communication
12	API Rate Limiting	Controlling API traffic

Level 4 — Database Architecture

Intermediate

This is one of the most important HLD topics.

No.	Topic	Concepts to cover
1	Database Selection	SQL vs NoSQL
2	Database Schema Design	Tables and relationships
3	Database Indexing	B-tree and indexing
4	Database Normalization	1NF, 2NF, 3NF
5	Database Denormalization	Improving read performance
6	Database Replication	Primary-replica architecture
7	Database Sharding	Horizontal partitioning
8	Database Partitioning	Range, hash and list
9	Read Replicas	Scaling read operations
10	Connection Pooling	Managing connections
11	Transactions	ACID properties
12	Database Consistency	Strong and eventual consistency

Level 5 — Scalability and Performance

Intermediate

Understand how to design applications that support increasing users and requests.

No.	Topic	Concepts to cover
1	Scalability Fundamentals	Scaling concepts
2	Vertical Scaling	Increasing machine resources
3	Horizontal Scaling	Adding more machines

4	Stateless Architecture	Stateless services
5	Stateful Architecture	Managing application state
6	Load Balancing	Traffic distribution
7	Caching Architecture	Cache placement
8	Cache Strategies	Cache-aside, write-through
9	Cache Eviction	LRU, LFU
10	CDN Architecture	Global content delivery
11	Database Optimization	Read/write optimization
12	Capacity Planning	Estimating resources
13	Bottleneck Analysis	Identifying performance limitations

Level 6 — Distributed Systems Architecture

Intermediate to Advanced

This level is extremely important for system design interviews.

No.	Topic	Concepts to cover
1	Distributed Systems Fundamentals	Multiple interconnected machines
2	CAP Theorem	Consistency, Availability, Partition Tolerance
3	PACELC Theorem	Latency and consistency trade-offs
4	Consistency Models	Strong, eventual, causal
5	Distributed Transactions	Two-phase commit
6	Saga Pattern	Managing distributed transactions
7	Event-Driven Architecture	Event-based communication
8	Message Queues	Producers and consumers
9	Event Streaming	Apache Kafka

10	Eventual Consistency	Data synchronization
11	Fault Tolerance	Handling failures
12	Circuit Breaker	Preventing cascading failures
13	Retry Mechanisms	Exponential backoff
14	Idempotency	Safe repeated operations
15	Distributed Locking	Coordinating concurrent operations

Level 7 — Reliability, Security and Monitoring

Advanced

No.	Topic	Concepts to cover
1	High Availability	Designing highly available systems
2	Fault Tolerance	Failure recovery
3	Disaster Recovery	Backup and restoration
4	Multi-Region Architecture	Geographical distribution
5	Replication Strategies	Synchronous and asynchronous
6	Authentication Architecture	Identity verification
7	Authorization Architecture	Access control
8	Encryption	At rest and in transit
9	Observability	Logs, metrics and traces
10	Monitoring	Infrastructure and application monitoring
11	SLI	Service Level Indicators
12	SLO	Service Level Objectives
13	SLA	Service Level Agreements
14	Chaos Engineering	Testing failure scenarios

Level 8 — Advanced Architectural Patterns

Advanced

These concepts are useful when designing complex production applications.

No.	Topic	Concepts to cover
1	Layered Architecture	Presentation, business and data layers
2	Hexagonal Architecture	Ports and adapters
3	Clean Architecture	Separation of concerns
4	Event Sourcing	Storing events as the source of truth
5	CQRS	Command Query Responsibility Segregation
6	Strangler Pattern	Gradual migration from monolith
7	Sidecar Pattern	Supporting service components
8	Bulkhead Pattern	Isolating failures
9	Backend for Frontend	Dedicated backend per client type
10	Service Mesh	Service-to-service management
11	Multi-Tenant Architecture	Supporting multiple customers
12	API Composition	Combining multiple service responses

Level 9 — AI and ML High-Level System Design

AI Engineer Priority

This is where you apply HLD to your actual career domain.

No.	Topic	Concepts to cover
1	ML System Architecture	End-to-end ML architecture
2	Data Ingestion Architecture	Batch and streaming
3	Data Processing Pipelines	ETL and ELT

4	Feature Store Architecture	Feature management
5	Model Training Architecture	Distributed training
6	Model Serving Architecture	Online and batch inference
7	Model Registry	Model version management
8	Model Monitoring	Model performance monitoring
9	Data Drift Architecture	Detecting data changes
10	Model Retraining Architecture	Automated retraining
11	GPU Infrastructure	GPU resource management
12	Distributed Inference	Multiple inference servers
13	Model Caching	Reusing inference results
14	Model Deployment	Canary and blue-green deployments

Level 10 — Generative AI and LLM System Architecture

Highest Priority for Modern AI Roles

No.	Topic	Concepts to cover
1	LLM Application Architecture	Overall LLM application design
2	RAG Architecture	Retrieval-Augmented Generation
3	Document Ingestion Pipeline	Parsing and processing
4	Chunking Architecture	Document segmentation
5	Embedding Architecture	Embedding generation
6	Vector Database Architecture	Vector storage and retrieval
7	Hybrid Search Architecture	Keyword and semantic retrieval
8	Reranking Architecture	Improving retrieval relevance
9	LLM Gateway	Centralized LLM access

10	LLM Load Balancing	Distributing model requests
11	LLM Caching	Semantic and response caching
12	Prompt Management	Prompt templates and versions
13	Agentic AI Architecture	Agents and tool calling
14	Multi-Agent Architecture	Agent orchestration
15	LLM Observability	Tracing and evaluation
16	LLM Guardrails	Input and output validation
17	LLM Security	Prompt injection and data leakage
18	GPU Inference Architecture	vLLM, batching and quantization
19	Token and Cost Optimization	Reducing inference expenses
20	Multi-Model Architecture	Routing between different LLMs

HLD Interview Practice — Design Problems

After learning the concepts, practise these architecture problems.

Difficulty	**Problem**
Beginner	Design a URL Shortener
Beginner	Design a File Storage System
Beginner	Design a Notification System
Beginner	Design a Rate Limiter
Intermediate	Design WhatsApp
Intermediate	Design YouTube
Intermediate	Design an E-commerce Platform
Intermediate	Design an Online Payment System
Intermediate	Design a Distributed Logging System

Advanced	Design a Real-Time Analytics Platform
Advanced	Design a Recommendation Engine
Advanced	Design a Fraud Detection System
AI-specific	Design ChatGPT
AI-specific	Design a Scalable RAG System
AI-specific	Design an AI Code Reviewer
AI-specific	Design an AI Agent Platform
AI-specific	Design a Production ML Pipeline

Recommended learning order

Although I have listed 10 levels, you do not need to master all advanced topics before starting AI-specific system design.

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